

Notes: Dewatripont & Tirole (JPE, 2005)

Modes of Communication

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1 Big picture

- **Question.** Why do some attempts at communication succeed, while others collapse into cheap talk, rubber-stamping, distrust, or information overload?
- **Core idea.** Communication requires costly effort by both sides. The sender must formulate, explain, and adapt the message; the receiver must pay attention, decode, and assimilate it.
- **Main departure from standard models.** The paper does not treat information as either costless hard information or costless cheap talk. Instead, information can be made more or less “hard” depending on costly communication and absorption efforts.
- **Modeling strategy.** Communication is modeled as a moral-hazard-in-teams problem: successful transmission requires both sender effort x and receiver effort y .
- **Main friction.** Even if the parties are aligned, each side privately bears its own communication cost but partly benefits from the other side’s effort. Hence equilibrium effort can be too low or collapse completely.
- **Decision regimes.** The paper distinguishes **executive decision making**, where the receiver must understand the message to implement action A , and **supervisory decision making**, where the receiver can rubber-stamp without understanding.
- **Surprising result.** More congruence between sender and receiver can reduce issue-relevant communication under supervisory decision making: the sender may rely on trust or authority and stop explaining.
- **Cues.** Communication may also involve cues about the sender’s credibility, expertise, or trustworthiness. Cues can either complement or substitute for issue-relevant messages.
- **Dynamics.** First impressions matter. Early communication success can raise beliefs about sender motivation and induce more effort later; early failure can trigger cognitive closure.
- **Economic takeaway.** Effective communication is not merely about preferences or information ownership. It is also about incentives to explain, incentives to listen, and the mode of decision making.

2 Incremental contribution of the paper

- **Beyond cheap talk.** Cheap-talk models focus on whether a sender’s message is credible given preference misalignment. Dewatripont and Tirole add the idea that even credible information may fail because sender and receiver do not exert enough costly communication effort.
- **Beyond hard information.** Disclosure models often assume that verifiable information can be transmitted and understood costlessly. This paper endogenizes how hard the information becomes: hardening information requires costly explanation and costly absorption.
- **Communication as joint production.** The paper imports the logic of moral hazard in teams into communication. A message is not simply sent; it is jointly produced by sender exposition and receiver attention.
- **Authority and rubber-stamping.** The paper explains why high trust or high congruence can paradoxically reduce the transmission of substantive information. When the receiver is willing to rubber-stamp, the sender may stop communicating issue-relevant content.
- **Dual-mode communication.** The paper formally distinguishes issue-relevant information from cues about the source. This connects economic communication theory with social psychology’s distinction between central/issue processing and peripheral/source cues.
- **Path dependence.** The paper shows how early success or failure can shape later communication. A first success signals sender motivation and encourages continuation; a first failure can make the receiver stop listening.
- **Information overload.** The framework can also rationalize strategic information overload: a sender may bury the receiver in information to make scrutiny harder while avoiding the appearance of concealment.

Interpretation: The incremental contribution is conceptual and theoretical. The paper creates a bridge between economic models of incentives and social-psychology evidence on persuasion, attention, elaboration, source credibility, and first impressions.

3 Model primitives and measurement

3.1 Players, action, and payoffs

There are two players:

- a sender S , who has information and wants action A to be chosen;
- a receiver R , who decides whether to choose action A or keep the status quo.

The status quo gives both players payoff zero. If action A is implemented, the sender receives

$$s > 0.$$

The receiver's payoff is uncertain:

$$r \in \{r_H, r_L\}, \quad r_H > 0 > r_L.$$

Let

$$\alpha = \Pr(r = r_H)$$

measure congruence or alignment between sender and receiver. The receiver would choose A without further information if

$$\alpha \geq \alpha^* \equiv \frac{-r_L}{r_H - r_L}.$$

Interpretation: α is the probability that the sender's preferred action is also good for the receiver. A high α means the receiver is more likely to trust the sender's recommendation.

3.2 Communication technology

Successful communication requires costly effort by both parties. The sender chooses effort x and the receiver chooses effort y . In the baseline model,

$$p(x, y) = xy.$$

The sender's effort cost is $S(x)$ and the receiver's effort cost is $R(y)$, both increasing and convex. The baseline technology implies strategic complementarity:

$$\frac{\partial^2 p(x, y)}{\partial x \partial y} > 0.$$

Interpretation: The sender's marginal return to explaining rises when the receiver listens harder. The receiver's marginal return to listening rises when the sender explains better. This is why communication can display virtuous circles or complete breakdowns.

3.3 Decision modes

The paper distinguishes two decision regimes.

- **Executive decision making.** The receiver can implement action A only if he understands the message. Communication is technologically necessary for implementation.
- **Supervisory decision making.** The receiver can choose A even without understanding the message. Communication only helps him decide whether to rubber-stamp.

Interpretation: The difference is central. Under executive decision making, issue-relevant communication remains necessary even when interests are aligned. Under supervisory decision making, high congruence may allow the sender to rely on authority and stop explaining.

3.4 Effort costs and communication-enabling parameters

The paper also considers a communication-enabling parameter v that lowers effort costs:

$$S = S(x, v), \quad R = R(y, v),$$

with

$$S_{xv} \leq 0, \quad R_{yv} \leq 0.$$

Examples include pedagogical talent, receiver absorptive capacity, shared language, common culture, or common references.

Interpretation: When v falls, communication becomes harder. The result may be a gradual decline in both efforts or a discrete communication breakdown if setup costs are present.

3.5 Cue versus issue-relevant information

The model later adds cues. Issue-relevant information tells the receiver whether action A is good. Cues tell the receiver whether the sender is credible, competent, or trustworthy.

The congruence parameter becomes uncertain:

$$\alpha = \gamma \bar{\alpha} + (1 - \gamma) \underline{\alpha}.$$

A cue reveals whether the sender is the high-credibility type $\bar{\alpha}$ or the low-credibility type $\underline{\alpha}$.

Interpretation: Cues are not about the project itself. They are about whether the receiver should believe that the sender's recommendation is worth processing.

4 Models and theoretical specifications

4.1 Baseline collaborative communication: sender does not know receiver's payoff

When $\alpha < \alpha^*$, the receiver will not choose A unless he understands that A is good. The payoffs are

$$U_S(x, y) = xy\alpha s - S(x),$$

$$U_R(x, y) = xy\alpha r_H - R(y).$$

An interior equilibrium satisfies

$$S'(x^*) = y^* \alpha s, \tag{1}$$

$$R'(y^*) = x^* \alpha r_H. \tag{2}$$

Interpretation: The sender explains more when the receiver's effort is higher, when the sender's payoff s is higher, and when congruence α is higher. The receiver listens more when the sender's effort is higher, the receiver's upside r_H is higher, and congruence is higher.

4.2 Communication breakdown

The zero-effort outcome

$$x^* = y^* = 0$$

is always an equilibrium. When both parties anticipate that the other will not try, neither side wants to bear the setup cost of communication.

Interpretation: This is the moral-hazard-in-teams problem. Communication is jointly produced, but each side's effort is privately costly. Mutual pessimism can become self-fulfilling.

4.3 High congruence and real authority

When $\alpha > \alpha^*$, the receiver is ex ante willing to choose A . Under supervisory decision making, the sender may choose

$$x^* = 0,$$

and the receiver may choose

$$y^* = 0.$$

This is the real-authority or rubber-stamping outcome.

Interpretation: More alignment can reduce substantive communication. If the receiver trusts the sender enough to approve the action, the sender has no reason to spend effort explaining details that might only create objections.

Proposition 1: supervisory versus executive decision making

Under supervisory decision making, greater congruence can lead to a breakdown of issue-relevant communication because the sender can count on the receiver to rubber-stamp. Under executive decision making, the receiver must understand in order to implement, so both payoffs increase monotonically with congruence.

4.4 Sender knows the receiver's payoff

If the sender knows whether $r = r_H$ or $r = r_L$, she does not exert effort when $r = r_L$. Conditional on $r = r_H$, interior efforts satisfy

$$R'(y^*) = x^* \alpha r_H, \tag{3}$$

$$S'(x^*) = y^* s. \tag{4}$$

When communication fails under supervisory decision making, the receiver updates negatively. A communication equilibrium requires the receiver not to choose A after failure:

$$\frac{\alpha(1 - x^* y^*)}{1 - \alpha x^* y^*} r_H + \frac{1 - \alpha}{1 - \alpha x^* y^*} r_L \leq 0. \tag{5}$$

The receiver must also prefer exerting effort to not exerting effort:

$$\alpha x^* y^* r_H - R(y^*) \geq \alpha r_H + (1 - \alpha) r_L. \tag{6}$$

Interpretation: When the sender is informed about receiver payoffs, silence itself becomes informative. Failure to communicate can signal that the project is bad for the receiver, sustaining a communication equilibrium even at higher congruence.

Proposition 2: informed sender

When the sender knows the receiver's payoff, communication may be sustained because failure to communicate carries bad news. Under supervisory decision making, both a real-authority equilibrium and an issue-relevant communication equilibrium can coexist over an intermediate congruence region.

4.5 Cues and the dual mode of communication

Cues reveal information about sender credibility. If cues are cheap, the receiver optimally processes the cue first and then decides how much issue-relevant effort to exert.

Proposition 3: cues can crowd in or crowd out issue-relevant communication

Bad-news cues about sender credibility can crowd out issue-relevant communication. Good-news cues can enhance issue-relevant communication if they raise credibility without inducing rubber-stamping. Very favorable cues can eliminate issue-relevant communication by giving the sender real authority.

Interpretation: Cues and issue-relevant messages are neither always complements nor always substitutes. Their relationship depends on whether the cue moves the receiver into distrust, active scrutiny, or rubber-stamping.

4.6 Strategic substitutability

The baseline assumes $p(x, y) = xy$, so efforts are strategic complements. The paper also considers

$$p = p(x + y), \quad p' > 0, \quad p'' < 0,$$

where the cross-partial is negative and efforts are strategic substitutes.

Proposition 4: robust total-effort results

With $p(x + y)$, the individual effort comparative statics need not match the baseline, but total communication effort still rises with congruence and stakes until real authority induces the breakdown of issue-relevant communication. Favorable cues raise total effort unless they create rubber-stamping; unfavorable cues reduce it.

Interpretation: The qualitative results do not depend only on strategic complementarity. Even if the sender and receiver can substitute for each other's effort, the same broad forces govern total communication.

4.7 Sequential communication and first impressions

Under pure sequentiality, the receiver observes the sender's effort and chooses a reaction $y = Y(x)$ satisfying

$$R'(Y(x)) = x\alpha r_H.$$

The sender becomes a Stackelberg leader and solves

$$\max_x \{xY(x)\alpha s - S(x)\}.$$

The first-order condition is

$$S'(\hat{x}) = [Y(\hat{x}) + Y'(\hat{x})\hat{x}]\alpha s.$$

Interpretation: The sender now has an additional incentive to explain clearly because doing so induces the receiver to listen harder. Sequentiality raises communication effort in the complementarity case.

4.8 Multistage communication

In the two-stage model, first-stage success has probability

$$p_1(x_1, y_1) = x_1 y_1.$$

At stage 2, success probabilities depend on whether stage 1 succeeded:

$$p_2(x_2, y_2) = M + x_2 y_2 \quad \text{after success,}$$

$$p_2(x_2, y_2) = m + x_2 y_2 \quad \text{after failure,}$$

where $M \geq m$.

The sender's payoff is private information:

$$s = s_H \text{ with probability } \rho, \quad s = s_L \text{ with probability } 1 - \rho.$$

After first-stage success or failure, the receiver updates beliefs to $\rho'(S)$ or $\rho'(F)$.

Proposition 5: first impressions matter

A successful first-stage communication signals a motivated sender: $\rho'(S) > \rho > \rho'(F)$. Communication is more effective at stage 2 after success than after failure. First-stage efforts are higher because early success improves later communication prospects. Failure is more likely to trigger cognitive closure.

Interpretation: The model formalizes path dependence in communication. Early success does not only transmit material; it changes beliefs about the sender's type and therefore changes the incentives to continue listening.

4.9 Strategic information overload

The paper also discusses cases in which more information can hurt the receiver. A sender may supply excessive detail to make scrutiny difficult while avoiding the appearance of concealment.

Interpretation: Strategic overload is most natural under supervisory decision making. The sender hopes the receiver will rubber-stamp, but cannot provide too little information without looking suspicious.

5 Identification and theoretical design

This paper is a theoretical paper, not an empirical identification exercise. The analogue of identification is the way the model isolates mechanisms by comparing decision regimes, information structures, and communication technologies.

5.1 What the model isolates

- **Moral hazard in teams.** Communication requires two costly efforts, so breakdown can occur even with aligned preferences.
- **Preference congruence.** The parameter α isolates how trust affects communication incentives.

- **Decision mode.** Comparing executive and supervisory decision making isolates the role of rubber-stamping.
- **Sender information.** Comparing whether the sender knows r isolates the signaling value of silence or failure.
- **Cues.** Adding credibility information separates source-based persuasion from issue-based persuasion.
- **Timing.** Sequential and multistage variants isolate first-impression and path-dependence effects.

Interpretation: The paper’s theoretical design is comparative. It changes one communication environment at a time and shows how equilibrium effort and knowledge transfer respond.

5.2 Why executive versus supervisory decision making is central

In executive decision making, communication is technologically necessary. In supervisory decision making, it is informationally useful but not necessary. Therefore:

$$\begin{aligned} \text{executive mode} &\Rightarrow \text{undercommunication problem,} \\ \text{supervisory mode} &\Rightarrow \text{undercommunication or rubber-stamping.} \end{aligned}$$

Interpretation: This distinction explains why trust is double-edged. Trust can encourage collaboration, but if the receiver can act without understanding, trust can undermine substantive explanation.

5.3 Why the sender’s knowledge matters

If the sender does not know whether A is good for the receiver, communication failure conveys little information. If the sender does know, failure may reveal that the project is bad.

$$\begin{aligned} \text{sender uninformed about } r &\Rightarrow \text{failure is uninformative,} \\ \text{sender informed about } r &\Rightarrow \text{failure may be bad news.} \end{aligned}$$

Interpretation: This is why the same communication technology can produce different equilibria depending on what the sender knows about the receiver’s payoff.

5.4 How cues identify source credibility effects

The cue extension separates two channels:

$$\begin{aligned} \text{issue-relevant message} &\Rightarrow \text{information about action } A, \\ \text{cue} &\Rightarrow \text{information about sender credibility.} \end{aligned}$$

Interpretation: This decomposition makes precise the social-psychology idea that receivers sometimes process the argument and sometimes process the source. The model shows when each mode dominates.

5.5 Main limitations of the theoretical design

- Communication effort is reduced to one-dimensional choices x and y .
- The baseline probability xy is stylized.
- Preferences and payoffs are simple binary objects.
- The model has no empirical calibration.
- Interactive communication, repeated clarification, and multiple senders are mostly left for future work.

Interpretation: These simplifications are deliberate. They make it possible to show cleanly why communication may collapse, why cues matter, and why first impressions generate path dependence.

6 Tables and figures

6.1 Figure 1: Payoffs under supervisory and executive decision making

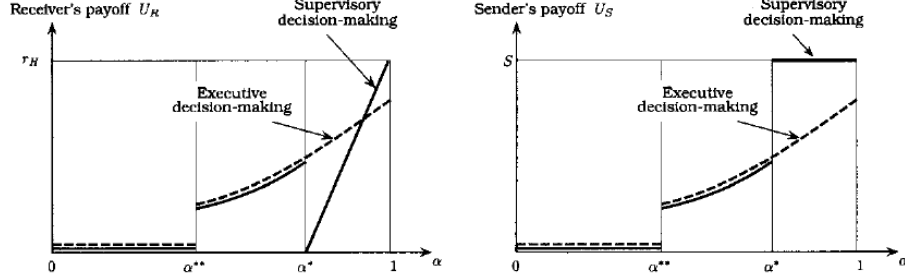


Figure 1: Receiver's and sender's payoffs under supervisory and executive decision making. The solid curves correspond to supervisory decision making, and the dashed curves correspond to executive decision making. The horizontal axis is congruence α . The figure shows that communication can be absent at low congruence, active in an intermediate range, and may collapse again under supervisory decision making when the sender obtains real authority.

Read:

- The left panel reports the receiver's payoff U_R as congruence α increases.
- The right panel reports the sender's payoff U_S as congruence α increases.
- For low congruence, communication may break down because neither side expects enough benefit from effort.
- In the intermediate region, communication occurs and both sides exert costly effort.
- Under executive decision making, payoffs rise more smoothly because action A requires successful understanding.
- Under supervisory decision making, the sender can eventually rely on rubber-stamping. At high congruence, issue-relevant communication disappears.
- The receiver can be worse off when congruence becomes too high because the sender stops explaining.

Economic message: The figure summarizes the paper's most important counterintuitive result: alignment improves communication only up to a point. When decision authority allows rubber-stamping, high trust can reduce substantive information transmission.

Interpretation: The discontinuity under supervisory decision making is not a graphical accident. It reflects a change in communication mode. Before the threshold, the sender must persuade through issue-relevant communication. After the threshold, the sender has real authority, so communication effort falls to zero. This is why the receiver's payoff can fall even though the sender's payoff rises.

6.2 Model result box: Proposition 1 - congruence and communication mode

Interpretation

Proposition 1 explains why congruence has different effects depending on the decision mode. In executive decision making, congruence raises the returns to both sender and receiver effort because understanding is necessary for implementation. In supervisory decision making, congruence first helps communication but then creates an authority region in which the receiver rubber-stamps and the sender no longer explains.

6.3 Model result box: Proposition 3 - cues versus issue-relevant communication

Interpretation

Cues are not always substitutes for arguments. A negative cue lowers credibility and can shut down issue-relevant effort. A moderately positive cue can make the receiver more willing to process the issue-relevant message, increasing effort. A very positive cue can eliminate issue-relevant effort because the receiver trusts the sender enough to approve without scrutiny.

6.4 Model result box: Proposition 5 - first impressions and cognitive closure

Interpretation

The multistage model shows that communication has history dependence. Early success signals sender motivation and increases later communication effort. Early failure lowers posterior beliefs and can lead the receiver to stop listening. This explains why first impressions can create virtuous circles or permanent breakdowns in communication.

7 Short summary

- The paper models communication as a costly joint production process.
- Successful communication requires both sender exposition effort and receiver absorption effort.
- Even aligned parties may undercommunicate because each side privately bears effort costs.
- Under supervisory decision making, high congruence can produce rubber-stamping and eliminate issue-relevant communication.
- Under executive decision making, the receiver must understand, so greater congruence tends to improve communication.
- If the sender knows the receiver's payoff, failure to communicate can itself signal bad news.
- Cues about sender credibility can either crowd in or crowd out issue-relevant communication.
- Sequential communication gives the sender stronger incentives to explain because exposition affects receiver effort.
- First-stage success can signal sender motivation and increase later communication; early failure can cause cognitive closure.
- Strategic information overload can arise when a sender wants the receiver to rubber-stamp while avoiding the appearance of concealment.

8 Conclusion

8.1 Core conclusion

Dewatripont and Tirole develop a theory in which communication is not costless message transmission. It is a joint activity in which the sender must explain and the receiver must absorb. Because both efforts are costly and privately chosen, communication may be too weak or may collapse entirely.

8.2 Economic intuition

The intuition is that it takes two to communicate. A clear lecture is wasted if the listener does not pay attention, and a motivated listener cannot learn much from a poorly prepared speaker. Each side's effort raises the productivity of the other side's effort. This creates strategic complementarities, coordination problems, and multiple equilibria.

8.3 Main conceptual contribution

The paper's main contribution is to integrate three forces usually studied separately: preference alignment, costly attention, and source credibility. It shows that trust does not simply improve communication. When the receiver can rubber-stamp, trust may reduce communication because the sender no longer needs to persuade with substantive content.

8.4 Implications for organizations

In organizations, the model explains why boards, managers, experts, and subordinates may fail to communicate even when they share broad goals. It also explains why high-status insiders may give less explanation, why recipients may rely on credentials rather than content, and why early interactions can shape future communication paths.

8.5 Implications for education and persuasion

The model has natural applications to teaching, training, advertising, political persuasion, and expert advice. Effective communication requires matching exposition to the receiver's absorptive capacity and giving the receiver incentives to process the message.

8.6 Limitations and future research

The paper leaves several important issues open. Communication is often interactive, and receivers can ask questions or request clarification. Multiple senders can compete or interfere. Messages can be multidimensional. Organizations can design incentives, routines, or authority structures to mitigate moral hazard in communication. The paper's framework provides a starting point for these extensions.

8.7 Takeaway

The central lesson is that communication failures are not only failures of honesty or information availability. They are also failures of effort, attention, credibility processing, and institutional design. Good communication requires incentives on both sides of the conversation.